# INTRODUCTION OF PYTHON

**Web Framework Using Python** is a comprehensive course designed to help learners understand how modern web applications are developed using Python-based web frameworks. The course starts from the basics of web development and gradually introduces backend programming concepts, making it suitable for learners who want to build dynamic and interactive web applications.

Students will first learn the fundamentals of **web applications, client-server architecture, HTTP requests and responses, URLs, web servers, and frontend-backend communication**. After understanding these concepts, the course introduces the **Flask framework**, including installation, project structure, application creation, routing, URL handling, request methods, and response handling.

The course then progresses to **HTML templates and Jinja2**, where students learn how to create dynamic web pages by connecting frontend templates with backend Python code. They will also learn how to create and process **HTML forms**, validate user input, handle sessions and cookies, and manage application data.

As learners progress, the course introduces **database integration using MongoDB**, including connecting Flask applications to MongoDB, creating collections and documents, performing CRUD operations, and retrieving and displaying database information through web pages. Students will also learn important concepts such as **user registration, login, authentication, authorization, and role-based access control**.

The later part of the course focuses on building more complete applications by combining **frontend, Flask backend, and database components**. Students will practice organizing applications, handling errors, validating data, and creating secure and user-friendly web interfaces. Finally, learners will apply everything they have learned by developing a **practical Flask-based web application**.



## 🎯 What You Will Learn

By completing this course, students will be able to:

- Understand how web applications work
- Learn client-server architecture and HTTP basics
- Set up Python and Flask development environments
- Create Flask applications from scratch
- Understand Flask project structure
- Create routes and handle URLs
- Work with GET and POST requests
- Create dynamic HTML pages using Jinja2
- Build and process HTML forms
- Validate and handle user input
- Use sessions and cookies
- Connect Flask applications with MongoDB
- Perform CRUD operations
- Implement user authentication and authorization
- Create role-based web applications
- Handle errors and application exceptions
- Connect frontend, backend, and database components
- Develop a complete practical web application

## 📚 Learning Path

**Basics → Web Concepts → Flask → Routing → Templates → Forms → Database → CRUD → Authentication → Role-Based Access → Project**

This can be used as the **main Overview/Introduction page** when a student opens the *Web Framework Using Python* course.